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To cite this article: Yongze Song (10 Sep 2026): Generalized covariate field (GCF): spatial-pattern and neighbourhood-distribution feature expansion improves geospatial prediction, International Journal of Geographical Information Science, DOI: [10.1080/13658816.2026.2729719](https://doi.org/10.1080/13658816.2026.2729719)

To link to this article: <https://doi.org/10.1080/13658816.2026.2729719>



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Published online: 10 Sep 2026.



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Generalized covariate field (GCF): spatial-pattern and neighbourhood-distribution feature expansion improves geospatial prediction

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ABSTRACT

Spatial prediction forms the basis for decision-making in Earth and environmental sciences, yet prediction accuracy is constrained by spatial heterogeneity and sparse and uneven sampling. Existing methods commonly respond by increasing algorithmic complexity while underusing the spatial information embedded in the covariates themselves. This study develops the generalized covariate field (GCF) to expand the covariate space by deriving two families of spatial features without increasing model complexity. First, spatial pattern features characterize local spatial structure, including spatial dependence, heterogeneity, geocomplexity, multiscale variation and local outlyingness, and are transferable across space. Second, neighbourhood distribution features summarize covariates through multiscale operators with geoscientific meaning. The resulting high-dimensional features are then selected through response-independent functional reduction and spatial-block stability selection. GCF is implemented in a vascular plant species-richness study across the Southwest Australian Floristic Region, and expands 12 covariates into 3276 candidate predictors. Results show that GCF improves prediction accuracy under both random and spatial-block validation with 8.2–21.1% higher spatial-block cross-validated R^2 than existing spatial learning models. The findings support the primacy of data hypothesis that enriching the spatial representation of data, rather than only increasing model structural complexity, can substantially improve prediction and provide a basis for data-centric geospatial modelling.

ARTICLE HISTORY

Received 27 June 2025

Accepted 1 September 2026

KEYWORDS

Geospatial prediction; generalized covariate field; feature expansion; spatial cross-validation; primacy of data

1. Introduction

Spatial prediction plays a fundamental role in Earth system science, supporting essential tasks such as climate modelling, biodiversity monitoring, ecosystem assessment and natural hazard forecasting (McKenzie and Ryan 1999, Brenning 2005). Such

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applications require not only accurate estimates at observed locations but also reliable predictions at unsampled or sparsely monitored areas (Wang *et al.* 2012, Meyer and Pebesma 2021). Advances in remote sensing and geospatial technologies greatly increased the availability and resolution of spatial data, yet more data alone do not guarantee higher prediction accuracy for complex systems (Chi *et al.* 2016). A key challenge remains the complex nature of spatial heterogeneity, driven by multiscale interactions, environmental gradients and spatially varying processes (Xia *et al.* 2023). Therefore, the effectiveness of spatial prediction depends not only on data availability but also on how well the underlying spatial structures in the data are identified and used in modelling.

Methods for spatial prediction can be broadly classified into three categories: spatial statistical models, spatially informed machine learning and integrated hybrid approaches. Spatial statistical models, such as kriging and its variants (e.g. ordinary kriging, universal kriging, regression kriging), provide best linear unbiased predictions by modelling spatial dependence through variograms. However, their performance can be sensitive to stationarity assumptions and sampling density, particularly in highly heterogeneous environments (Li and Heap 2014, Oliver and Webster 2014). Spatial autoregressive models and geographically weighted regression extend regression by incorporating spatial autocorrelation or spatially varying coefficients (Anselin 1995, Fotheringham and Brunsdon 1999). Machine learning algorithms such as random forests (RFs), support vector machines (SVMs) and gradient boosting show strong prediction performance when integrating spatial covariates (Rodriguez-Galiano *et al.* 2015). However, many machine learning models treat observations as independently distributed and fail to account for spatial distribution patterns, which limits their effectiveness in spatially structured environments (Du *et al.* 2020). Emerging approaches address this limitation by embedding spatial variables into the input space or adapting the learning algorithm to capture spatial dependencies (Fischer 1998, Goodchild and Li 2021). Despite these advances, challenges remain in systematically representing and learning from spatial characteristics in machine learning, particularly for prediction in unsampled or complex environments (Jemeljanova *et al.* 2024).

Spatial characteristics define the structure and function of spatial modelling and are essential for the representation, learning and prediction of spatial information (Yuan 2024, Liu *et al.* 2026). They refer to the intrinsic patterns, structures and relationships in spatial data that models learn to encode and generalize for prediction (Yuan and Hornsby 2007, Zhu *et al.* 2018, Yang *et al.* 2024). Spatial representations are not merely derived from neighboring data. Rather, they are learned expressions of meaningful spatial characteristics such as spatial dependence (Anselin 1995), heterogeneity (Wang *et al.* 2010, Goodchild and Li 2021), multiscale patterns (Ge *et al.* 2016, 2019), geocomplexity (Zhang *et al.* 2023), singularity (Cheng 2012), outliers (Aggarwal and Aggarwal 2017) and texture (Mohanaiah *et al.* 2013). For instance, geocomplexity, quantified using the spatial local complexity indicator, characterizes complex local spatial patterns while considering spatial neighbor dependence and can effectively explain spatial errors in machine learning and spatial learning (Zhang *et al.* 2023). Singularity, a property of nonlinear natural processes, produces fractal or multifractal patterns that

reflect geo-abnormalities and help detect extremes for improved prediction in irregular spatial regions (Cheng 2012).

Spatial heterogeneity is a property of the geospatial observation field, under which the local statistical properties of a variable vary with location rather than remaining constant across the domain. Variation of this kind takes two forms: local heterogeneity, in which those properties change continuously from place to place (Fotheringham and Brunsdon 1999), and stratified heterogeneity, in which the study area divides into strata whose internal variance is smaller than the variance of the space as a whole (Wang *et al.* 2010). In recent years, spatial learning models have evolved across multiple methodological categories to address spatial autocorrelation, heterogeneity and other spatial characteristics in prediction tasks. Classical spatial statistical models, such as spatial autoregressive models and geographically weighted regression, incorporate spatial dependencies or allow model parameters to vary locally based on geographic coordinates (Ver Hoef *et al.* 2018, Li *et al.* 2019). Dependence-based methods include heterogeneous autocorrelation models adapting ensemble weights based on local patterns (Zhang *et al.* 2024). Heterogeneity-focused models capture multiscale, stratified or mixed spatial variations (Zhang *et al.* 2022, Hu *et al.* 2025, Luo *et al.* 2026b); for example, the generalized heterogeneity model (GHM) characterizes intra- and inter-stratum variation to improve interpolation near stratum boundaries (Luo *et al.* 2023), while the geocomplexity indicator quantifies local spatial complexity relative to neighbor dependence, supporting error diagnosis in aspatial and spatial models (Zhang *et al.* 2023). Similarity-based approaches, such as geographical configuration similarity and the geographically optimal similarity model, improve prediction by identifying structurally similar observations rather than relying solely on proximity (Zhu *et al.* 2018, Song 2023, Zhao *et al.* 2024). Machine learning and hybrid spatial models embed spatial structure into nonlinear prediction algorithms (Aburas *et al.* 2019), while deep learning methods, including graph convolutional networks and spatial attention mechanisms, capture higher-order spatial relationships but remain limited in interpretability and computational scalability (Guo *et al.* 2019).

Beyond models that rely solely on covariate values at sampled locations, spatial prediction can use the covariate field, which captures the values and spatial structure of covariates in the neighbourhood surrounding each prediction location, including unsampled locations. Because covariates are often available across this neighbourhood even when the response is unobserved, their local distributions and spatial patterns provide information beyond point covariate values (Luo *et al.* 2026a). Summarizing this surrounding covariate information without increasing the number of response observations broadens the representation available to spatial learning and supports more reliable estimation at unsampled or sparsely monitored locations (Meyer and Pebesma 2021, Song 2022).

Despite recent advances, gaps remain in systematically quantifying and modelling spatial characteristics beyond the sampled locations. Many approaches remain model-centric, prioritizing algorithmic complexity rather than systematically examining the multiple types and scales of spatial structures embedded in the data. Variables are commonly used as basic predictors instead of as sources of deeper characteristics such as local variation, anisotropy or singularity, which constrains the ability to capture

the full complexity of heterogeneous environments. Response information is available only at sampled locations, which limits the use of the surrounding covariate field for prediction in unsampled or sparsely monitored regions. Although progress in local spatial indicators, graph-based techniques and similarity-based models broadened the field, a systematic approach for generating and integrating spatial pattern and neighbourhood distribution features from covariates at unsampled locations is still lacking (Song 2022).

The study develops the generalized covariate field (GCF) model to expand each covariate into two complementary sets of features across space for geospatial prediction. A spatial pattern operator derives spatial pattern features, such as spatial dependence, local heterogeneity, multiscale variation, geocomplexity and local outlyingness, and a neighbourhood distribution operator summarizes the distribution of covariates over multiscale neighbourhoods, yielding neighbourhood distribution features. Together with the original predictors, these form the three predictor categories of GCF and extend covariate information from the sampled points to unsampled areas, enhancing prediction where direct measurement is infeasible. By expanding the predictor space rather than the model structure, GCF demonstrates the primacy of data principle: deeper spatial data mining can yield greater improvements in prediction accuracy than further increases in model structural complexity. GCF is validated through a spatial simulation and an application predicting species richness in the Southwest Australian Floristic Region (SWAFR) using 12 environmental variables. Results show that GCF consistently improves model fit and reduces error compared to established spatial learning models, and is demonstrated as a robust approach for advancing geospatial prediction and multiscale spatial understanding.

2. Generalized covariate field

2.1. Concepts and definitions

The GCF is a spatial representation model that enriches each covariate at both sampling locations and around the prediction locations, using the covariate field rather than its point value alone to improve spatial learning and prediction. The term generalized refers to the use of the covariate field with the spatial structure of a covariate and the distribution of covariates over its neighbourhood, so that conventional predictor sets, namely the covariates alone, the covariates with spatial features, and the covariates with neighbourhood statistics, arise as special cases.

GCF organizes its predictors into three categories: the original predictors x , the spatial pattern features $\psi(x)$ and the neighbourhood distribution features Z_x . GCF is implemented through four main steps. A spatial pattern operator first generates the spatial pattern features to describe spatial dependence, heterogeneity, multiscale variation, geocomplexity and local outlyingness for each covariate, which are invariant and transferable across space. A neighbourhood distribution operator then summarizes covariate distributions over multiscale neighbourhoods of each prediction location using quantiles to produce neighbourhood distribution features (Song 2022). Variable selection is subsequently applied to the candidate predictor set formed from the three categories through a response-independent functional reduction followed by

spatial-block stability selection, and a spatial learning model is finally trained on the selected predictors. By expanding the predictor space around each location rather than the number of observations, GCF enables spatial models to learn from a richer covariate representation and enhances the generalizability and accuracy of predictions across diverse spatial contexts.

Spatial representation of data denotes the quantities supplied to a learner as predictors for each location, comprising the covariate values at that location together with quantities encoding the spatial structure of the covariate field in its neighbourhood. Under this definition, the predictor set $x(s)$ is the point-support representation, and the GCF feature vector $\Phi(s)$ is an enriched representation built from the same observations without adding response samples. Spatial pattern operators in GCF are neighbourhood functionals, so the features characterize local heterogeneity of the covariate field together with local dependence, multiscale variation and local outlyingness, and the operator set is open, since any functional of covariate values over a defined spatial support can be added to ψ without altering the construction. Three conditions govern the benefit of GCF. First, the response depends on the geographical context of a location rather than on its point covariate values alone. Second, covariates are observed on a support extending beyond the response sample locations, so that neighbourhood summaries carry information the response samples do not. Third, the neighbourhood radii span the scales at which the underlying process operates. The more strongly the local structure of a covariate field varies across the domain, the more information the GCF features carry, and the largest gains are expected where that variation is strongest.

2.2. Model

GCF is developed to characterize the geographical context surrounding each location by encoding the spatial structure of covariates and their distributions within local neighbourhoods. This enables spatial prediction to exploit spatial characteristics inherent to geographical processes, rather than relying solely on point covariate values. GCF is formalized through four steps, including spatial pattern feature construction, neighbourhood distribution feature construction, variable selection and spatial learning. For any location $s \in \mathcal{D}$, the GCF features are defined as follows. Let $\mathcal{S} \subset \mathcal{D}$ denote the sampled locations u , where both the covariates $x(u)$ and the response $y(u)$ are observed. Let $\mathcal{G} \subset \mathcal{D}$ denote the prediction grid, whose elements v are prediction locations where covariates $x(v)$ are available but the response is unobserved. The construction is therefore defined over locations with covariate support, including both sampled locations and prediction locations. Let x denote the original predictors. GCF predicts the response at a sampled location $u \in \mathcal{S}$ through

$$y(u) = f(\Phi(u)), \quad u \in \mathcal{S}, \quad (1)$$

where $\Phi(u) = [x(u), \psi(x)(u), \{Z_x(u; b, \tau)\}]$ is the GCF feature vector evaluated at each sampled location $u \in \mathcal{S}$. It comprises the original predictors x , the spatial pattern features $\psi(x)$ and the neighbourhood distribution features Z_x . Stacking these feature vectors over all sampled locations yields the GCF feature matrix $\mathbf{X}_{\text{GCF}}$. The learner f maps

$\Phi(u)$ to the response $y(u)$ and is fully replaceable. This study makes a methodological contribution to the construction of Φ , rather than to the choice of learner.

The spatial pattern operator ψ acts on the covariate field x within the neighbourhood $\mathcal{N}(s)$ of a location s , producing descriptors of local spatial structure, denoted by $\psi(x)(s)$. The neighbourhood distribution operator Z summarizes the distribution of covariate values within a multiscale neighbourhood $\mathcal{N}_b(s)$ through distributional summaries indexed by scale b and quantile level τ , producing the features $Z_x(s; b, \tau)$. The two operators are described in detail below.

2.2.1. Spatial pattern features

The first stage constructs spatial pattern features by applying the operator ψ to each original predictor. For each covariate x , the operator ψ produces an h -dimensional vector of spatial pattern features,

$$\psi(x)(s) = (\psi_1(x)(s), \psi_2(x)(s), \dots, \psi_h(x)(s)), \quad s \in \mathcal{D}, \quad (2)$$

where $\psi_i(x)(s)$ is the i th spatial pattern feature computed from the values of x within the neighbourhood $\mathcal{N}(s)$. The same set of h spatial pattern operators is applied independently to each original predictor.

Table 1 shows the 11 operators used to construct the spatial pattern features. These operators characterize complementary aspects of local spatial structure, including spatial dependence, local heterogeneity, distributional variation, geocomplexity, multiscale variation and spatial outlyingness. Spatial dependence is represented by a transferable LISA-type statistic (ψ_L), which measures local spatial association using locally standardized covariate values, and local Geary contrast (ψ_C). By removing broad-scale differences in level and variability, ψ_L focuses on local spatial patterns. Local heterogeneity and disorder are captured by log local variance (ψ_V) and rank-binned local entropy (ψ_E) (Batty 1974, Woodcock and Strahler 1987, Anselin 1995). Geocomplexity (ψ_G) integrates spatial variation with neighbour dependence, and multiscale structure is represented by the log scale-variance index (ψ_S) and local variogram exponent (ψ_T) (Moellering and Tobler 1972, Oliver and Webster 2014, Zhang *et al.* 2023). Local outlyingness is captured by positive and negative outlier strengths in both local- z and robust median forms (ψ_P and ψ_N) (Aggarwal and Aggarwal 2017, Ren *et al.* 2026). Together, these operators transform each covariate x into the spatial pattern feature vector $\psi(x)(s)$ to incorporate localized and multiscale spatial structure.

2.2.2. Neighbourhood distribution features

The second stage constructs neighbourhood distribution features by applying the operator Z to each covariate over multiscale neighbourhoods. This stage summarizes the distribution of covariate values surrounding each location using quantiles. Covariates are kept at their native resolution, so the construction does not require all covariates to be resampled to a common resolution.

For a covariate x , buffer radii $b \in \mathcal{B}$ define multiscale neighbourhoods $\mathcal{N}(s; b)$ around each location $s \in \mathcal{D}$. Within each neighbourhood, a distribution-summary operator Q_τ , taken here as the quantile at level τ , is applied to the covariate values at locations with covariate support (Song 2022),

Table 1. The 11 spatial pattern operators used in the generalized covariate field (GCF).

Category	Indicator	Definition
Spatial dependence	Transferable LISA-type statistic ψ_L	$\psi_L(s) = \bar{z}_s \cdot m^{-1} \sum_{j \in \mathcal{N}(s)} \bar{z}_j, \bar{z}_a = (x(a) - \mu_{a,r}) / \sigma_{a,r}, a \in \{s\} \cup \mathcal{N}(s)$ (adapted from Anselin 1995)
	Local Geary's c ψ_C	$\psi_C(s) = \sigma_x^{-2} \cdot m^{-1} \sum_{j \in \mathcal{N}(s)} (x(s) - x(j))^2$ (Anselin 1995)
Local heterogeneity	Log local variance ψ_V	$\psi_V(s) = \log(1 + (m-1)^{-1} \sum_{j \in \mathcal{N}(s)} (x(j) - \mu_s)^2)$ (Woodcock and Strahler 1987)
Distributional diversity	Rank-binned entropy ψ_E	$\psi_E(s) = -(\log K)^{-1} \sum_{k=1}^K p_k(s) \log p_k(s), p_k(s) = m^{-1} \sum_{j \in \mathcal{N}(s)} \mathbb{I}\{x(j) \in B_k\}$ (Batty 1974)
	Geocomplexity ψ_G	$\psi_G(s) = -\sigma_x^{-2} m^{-1} [\bar{\delta}_s \sum_{j \in \mathcal{N}(s)} w_{sj} \bar{\delta}_j] / \sum_{j \in \mathcal{N}(s)} w_{sj} \bar{\delta}_j \cdot \mathcal{N}(s) \cap \mathcal{N}(j) ^{-1} \sum_{i \in \mathcal{N}(s) \cap \mathcal{N}(j)} w_{si} w_{ij} \bar{\delta}_i, \bar{\delta}_a = x(a) - \bar{x}$ (Zhang et al. 2023)
Multiscale variation	Log scale-variance index ψ_S	$\psi_S(s) = \log(1 + \text{Var}_{\tau_h}\{\mu_{s,h}\}), \mu_{s,h} = C_h(s) ^{-1} \sum_{j \in C_h(s)} x(j)$ (Moellering and Tobler 1972)
	Local variogram exponent ψ_T	$\psi_T(s) = d \log \gamma_s(h) / d \log h, \gamma_s(h_k) = \{2n_k(s)\}^{-1} \sum_{(a,a') \in P_k(s)} (x(a) - x(a'))^2$ (Oliver and Webster 2014)
Local outlyingness	Local-z positive outlier strength ψ_P	$\psi_P(s) = \sum_{j \in \mathcal{N}(s)} z_j \mathbb{I}(z_j > \theta), z_j = (x(j) - \mu_s) / \sigma_s$ (Aggarwal and Aggarwal 2017, Ren et al. 2026)
	Local-z negative outlier strength ψ_W	$\psi_W(s) = \sum_{j \in \mathcal{N}(s)} z_j \mathbb{I}(z_j < -\theta), z_j = (x(j) - \mu_s) / \sigma_s$ (Aggarwal and Aggarwal 2017, Ren et al. 2026)
	Robust positive outlier strength ψ_P'	$\psi_P'(s) = \sum_{j \in \mathcal{N}(s)} r_j \mathbb{I}(r_j > \theta), r_j = (x(j) - \text{med}_s) / \text{MAD}_s$ (Aggarwal and Aggarwal 2017)
	Robust negative outlier strength ψ_W'	$\psi_W'(s) = \sum_{j \in \mathcal{N}(s)} r_j \mathbb{I}(r_j < -\theta), r_j = (x(j) - \text{med}_s) / \text{MAD}_s$ (Aggarwal and Aggarwal 2017)

Each operator summarizes covariate values within a local neighbourhood $\mathcal{N}(s)$ into a spatial pattern feature designed to capture transferable local spatial structure.

Here, $m = |\mathcal{N}(s)|$, and $\bar{x}$ and σ_x^2 are the global mean and variance of x .

For ψ_L , $\mu_{a,r}$ and $\sigma_{a,r}$ are the mean and standard deviation of x within $\mathcal{N}_r(a)$, the neighbourhood centred at a with normalization radius r .

For ψ_V , ψ_P and ψ_W , μ_s and σ_s are the local mean and standard deviation of x .

For ψ_E , B_k is the k th global rank bin.

For ψ_G , w_{ij} is the spatial weight between s and j .

For ψ_S , $C_h(s)$ is the grid cell of side length h containing s .

For ψ_T , $P_k(s)$ denotes the point pairs within $\{s\} \cup \mathcal{N}(s)$ in the k th distance bin, $n_k(s) = |P_k(s)|$, h_k is the mean pair distance in bin k , and the slope is estimated by weighted least squares using $n_k(s)$ as weights.

For ψ_P , ψ_W , ψ_P' and ψ_W' , θ is the outlier threshold; med_s and MAD_s in the robust forms are the local median and scaled median absolute deviation, respectively.

$$Z_x(s; b, \tau) = Q_\tau(\{x(w) : w \in \mathcal{N}(s; b)\}), \quad s \in \mathcal{D}, \quad (3)$$

where w indexes the locations with covariate support within $\mathcal{N}(s; b)$, and $\tau \in [0, 1]$ is the quantile level. Although quantiles are used in this study, Q_τ can be replaced by other distributional summaries, such as means, upper-tail statistics, threshold exceedances or entropy, without changing the general construction.

The combinations of buffer radii $b \in \mathcal{B}$ and quantile levels $\tau \in \mathcal{T}$ generate the neighbourhood distribution features, with one feature for each buffer–quantile combination. These features are combined with the original predictors and the spatial pattern features in the GCF feature vector $\Phi(s)$, which is evaluated at sampled locations $u \in \mathcal{S}$ for training and at prediction locations $v \in \mathcal{G}$ for prediction.

2.2.3. Variable selection

The candidate predictor set consists of the entries of the GCF feature vector $\Phi(s)$, including the original predictors $x(s)$, the spatial pattern features $\psi(x)(s)$ and the neighbourhood distribution features $Z_x(s; b, \tau)$. These derived features form a high-dimensional and strongly collinear candidate space because neighbouring buffers, adjacent quantile levels and related spatial operators often encode similar information. Variable selection is therefore used to obtain a compact subset of low-collinearity and spatially reproducible features. The procedure has two stages: response-independent functional reduction followed by spatial-block stability selection.

Functional reduction collapses redundant buffer and quantile sweeps into interpretable summaries for each covariate and scale band. The buffer set is partitioned into a fine band and a broad band, preserving a near–far contrast while reducing resolution-to-resolution redundancy. For the neighbourhood distribution features, each band is summarized by five functionals of the quantile curve: the median, interquartile range, lower tail, upper tail and skew descriptor. For the spatial pattern features, each scale-dependent operator is averaged within each scale band, while scale-invariant operators pass through unchanged. Because the reduction uses only covariate information, the resulting representation is response-independent and does not introduce response leakage. Each retained column preserves its covariate identity and feature category, namely original predictor, spatial pattern feature or neighbourhood distribution feature.

Spatial-block stability selection is then applied to the reduced derived-feature set. The original predictors are forced into all models and are exempt from selection, so derived features are selected only for information beyond that already represented by x . The study region is partitioned into contiguous square blocks. In each of M spatial-block resampling rounds, a fraction γ of blocks is sampled, a RF is fitted on the selected blocks, and the top k derived features ranked by impurity importance are marked as selected. Features are grouped by covariate and feature category. For a group g , the selection frequency is

$$\hat{p}_g = \frac{1}{M} \sum_{r=1}^M \mathbb{I} \left\{ \sum_{j \in g} \delta_{rj} > 0 \right\}, \quad (4)$$

where M is the number of spatial-block resampling rounds, and $\delta_{rj} = 1$ if feature j is selected in round r , and $\delta_{rj} = 0$ otherwise. Groups with $\hat{p}_g \geq \pi$ are retained, where π controls the trade-off between recall and stability. Each retained group contributes

one representative feature, chosen as the member with the highest mean selection frequency. The final predictor set is the union of the forced original predictors and these representative derived features.

2.2.4. Spatial modelling and prediction

The final stage trains a supervised learner using the predictors derived from the variable selection stage and applies the fitted model across the prediction locations. Let $\mathcal{A}$ denote the selected predictor set, including the forced original predictors and the selected representative features. The selected GCF feature vector at location s is

$$\Phi_{\mathcal{A}}(s) = (\Phi_j(s) : j \in \mathcal{A}), \quad (5)$$

where $\Phi_j(s)$ is the j th candidate feature in the GCF feature vector $\Phi(s)$. The training matrix is obtained by evaluating $\Phi_{\mathcal{A}}(u)$ at the sampled locations $u \in \mathcal{S}$, together with the observed responses $y(u)$.

The learner f is replaceable, and any suitable supervised model, such as a RF or a Cubist model, can be used in this stage. After fitting f on the sampled locations, prediction at an unknown location $v \in \mathcal{G}$ is given by

$$\hat{y}(v) = f(\Phi_{\mathcal{A}}(v)), \quad v \in \mathcal{G}. \quad (6)$$

The resulting predictions form the spatial prediction surface over the prediction locations with available covariate values.

2.3. Simulation study

2.3.1. Simulation design and GCF modelling

A spatial simulation experiment was conducted to evaluate the prediction performance of GCF. The simulation dataset comprises a regular 30×30 spatial grid containing one dependent variable y and three spatially distributed explanatory variables x_1 , x_2 and x_3 , as shown in Figure 1. The response is generated as a spatially autocorrelated Gaussian random field, and each explanatory variable is generated by adding spatially independent noise to that field.

GCF is applied to the three explanatory variables to construct three predictor categories: original predictors, spatial pattern features and neighbourhood distribution features. Using the 11 spatial pattern operators listed in Table 1, together with buffer radii $b = \{2, 4, 6\}$, the normalization radius $r = 4$ for ψ_L , and quantile levels $\tau = \{0, 0.1, \dots, 1\}$ for the neighbourhood distribution operator, each explanatory variable yields $1 + (9 \times 3 + 2) + (3 \times 11) = 63$ predictors. These comprise 1 original predictor, 29 spatial pattern features and 33 neighbourhood distribution features. Across the three explanatory variables, the GCF candidate set contains $3 \times 63 = 189$ predictors.

To reduce dimensionality and redundancy, the variable selection procedure is applied to the candidate predictor set. This consists of response-independent functional reduction followed by spatial-block stability selection with group voting, and is re-estimated within each training fold during cross-validation. Machine learning models, including RF, Cubist, GLMNET, linear regression model (LM), XGBoost, SVM and distance-weighted k -nearest neighbours (KNN), are used to model the relationship between the selected predictors and the response variable $y(u)$. Model performance is evaluated using fivefold

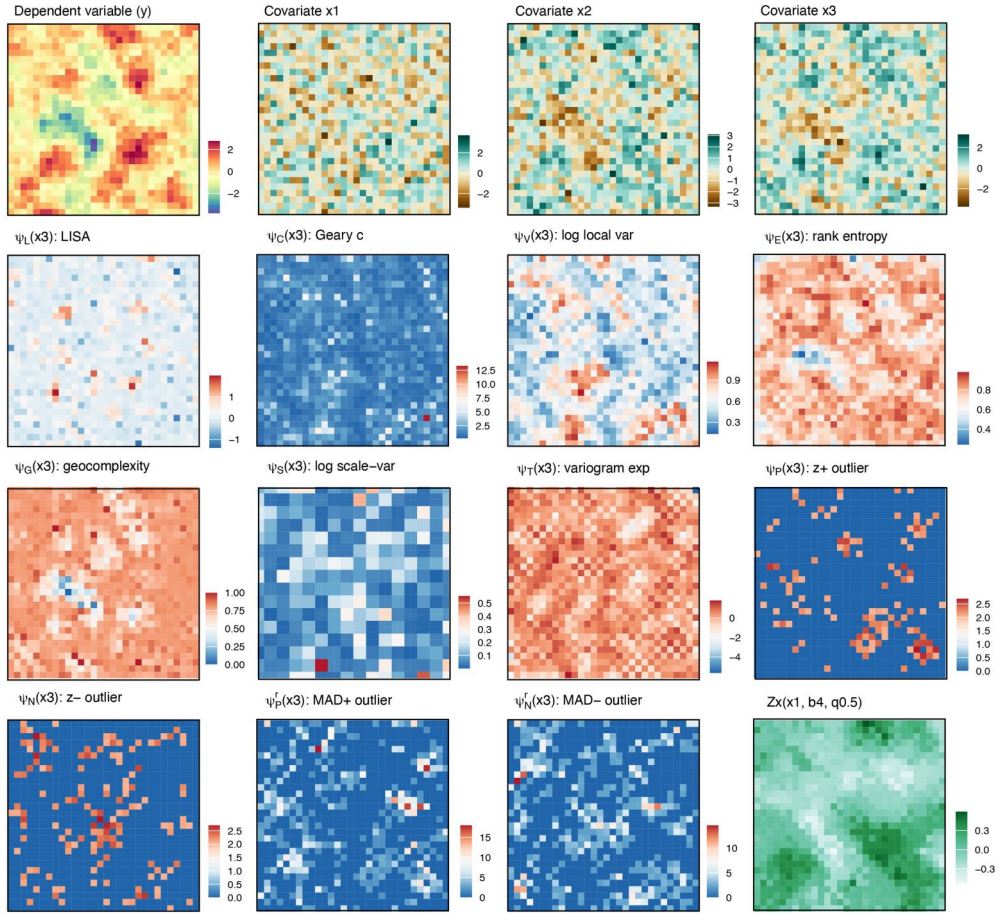


Figure 1. Spatial distributions of the simulated dependent and independent variables, along with selected representative examples of spatial pattern features for an explanatory variable and neighbourhood distribution features used in the GCF model.

random and spatial-block cross-validation. The validation metrics coefficient of determination (R^2) and root mean square error (RMSE) are calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (7)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (8)$$

where y_i is the observed value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the observed values in the validation fold and n is the number of validation samples.

2.3.2. Simulation results and evaluation

Table 2 shows the fivefold random and spatial-block cross-validation results for the simulation, comparing the covariate-only models with the GCF models among seven learners. Under spatial-block cross-validation, GCF achieves R^2 values ranging from

Table 2. Random and spatial-block cross-validation for the simulation study.

Model	Random CV						Spatial CV					
	R^2			RMSE			R^2			RMSE		
	Baseline	GCF	Imp. (%)	Baseline	GCF	Imp. (%)	Baseline	GCF	Imp. (%)	Baseline	GCF	Imp. (%)
LM	0.514	0.700	+36.0%	0.700	0.550	+21.4%	0.469	0.668	+42.4%	0.706	0.551	+22.0%
GLMNET	0.514	0.700	+36.2%	0.701	0.550	+21.4%	0.470	0.668	+42.1%	0.707	0.552	+22.0%
RF	0.457	0.697	+52.6%	0.740	0.552	+25.4%	0.419	0.657	+56.5%	0.741	0.561	+24.3%
Cubist	0.476	0.678	+42.3%	0.727	0.570	+21.6%	0.438	0.637	+45.6%	0.728	0.575	+21.0%
KNN	0.467	0.684	+46.5%	0.734	0.565	+23.0%	0.429	0.637	+48.5%	0.736	0.578	+21.5%
XGBoost	0.445	0.677	+52.1%	0.748	0.571	+23.7%	0.399	0.619	+55.2%	0.754	0.590	+21.8%
SVM	0.472	0.645	+36.7%	0.730	0.598	+18.0%	0.433	0.591	+36.6%	0.734	0.624	+15.0%

Baseline and GCF denote the covariate-only models and the GCF models, respectively; Imp. (%) is the relative improvement by GCF models compared with corresponding covariate-only models; bold values indicate the best cross-validation result in each column.

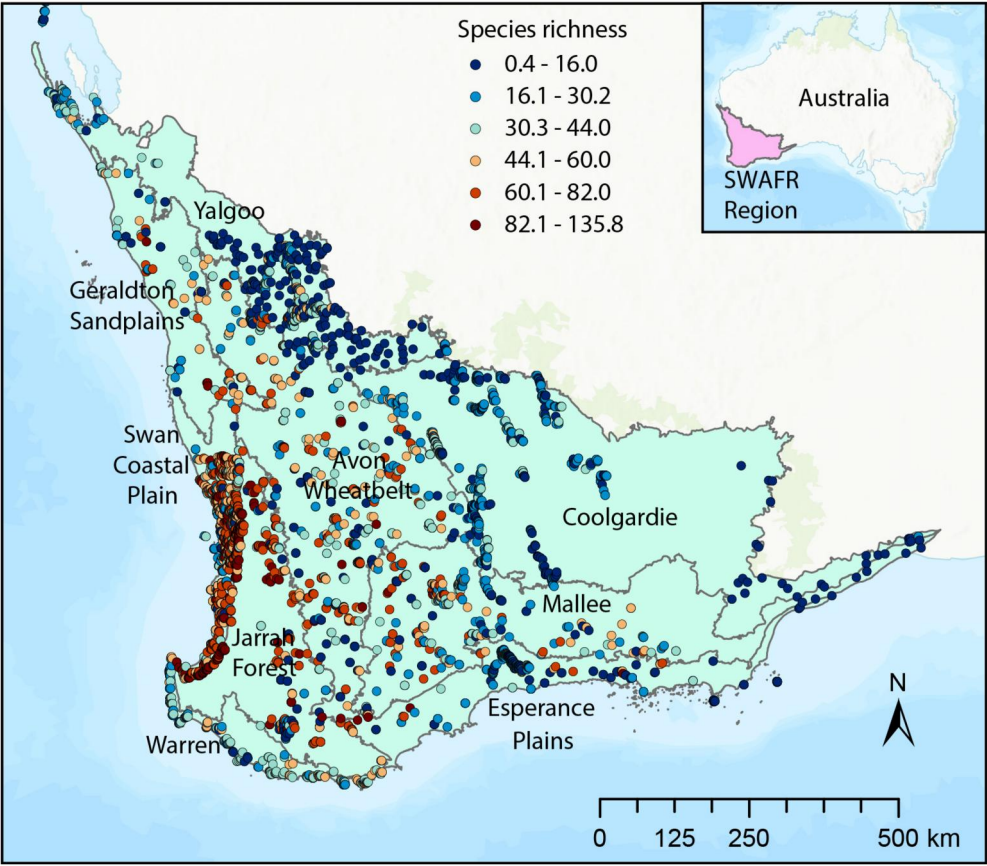


Figure 2. Spatial distribution of samples of species richness in the Southwest Australian Floristic Region (SWAFR) in Australia.

Table 3. Descriptions of variables used for the spatial prediction of species richness.

Category	Variable	Code	Description and unit
Geography	Elevation	Elevation	Elevation (m)
	Slope	Slope	Terrain slope (°)
Climate and environment	Precipitation	Precipitation	Total precipitation (mm)
	Short wave radiation	Radiation	Annual average short wave radiation (W/m ²)
	Distance to water	DistWater	Distance to the nearest water body (km)
	Distance to built-up areas	DistBuilt	Distance to artificial land or urban area (km)
Soil conditions	Soil N	SoilN	Soil total nitrogen content (%)
	Soil organic carbon	SoilC	Soil organic carbon content (%)
	Soil clay	SoilClay	Soil clay fraction (%)
	Soil depth	SoilDepth	Depth to impermeable layer from surface (m)
	Soil pH	SoilpH	Average soil pH value
	Soil bulk density	SoilBD	Dry soil mass per unit volume (g/cm ³)

0.591 for SVM to 0.668 for LM, compared with 0.399–0.470 for the covariate-only models. Among the learners, GCF-based LM has the lowest spatial RMSE with 0.551, compared with 0.706 for the covariate-only LM. The strong performance of GCF-based LM is consistent with the simulation design and the GCF representation, as the spatial pattern and neighbourhood distribution features encode much of the spatial structure

before model fitting with a comparatively simple mapping from predictors to response. Prediction with the covariates alone also favours linear modelling, with a spatial R^2 of 0.470 for LM compared with 0.420 for RF, which shows limited residual nonlinearity for tree-based learners to exploit. As the prediction accuracy of all seven GCF-based learners improves over their corresponding covariate-only models under both random and spatial-block cross-validation, the improvement is attributable primarily to the GCF representation rather than to any particular learning algorithm.

3. Case: GCF-based spatial prediction of biodiversity

3.1. Study area and data

The SWAFR, situated in the southwestern corner of Australia, is a globally significant biodiversity hotspot characterized by exceptional vascular plant richness, high levels of endemism, and strong ecological gradients (Monks *et al.* 2019). The region exhibits a Mediterranean climate, with cool, wet winters and hot, dry summers, and is underlain by ancient, highly weathered and nutrient-deficient soils influencing vegetation structure and function (Hopper and Gioia 2004, Brundrett 2021). Approximately, 7380 native vascular plant species and subspecies have been recorded in SWAFR, with half classified as endemic and more than 2500 species listed as of conservation concern (Hopper and Gioia 2004). The region is hydrologically defined by two major isohyet boundaries at 600 mm and 300 mm of annual precipitation, delineating distinct rainfall provinces, where the Transitional Rainfall Province between them has the highest floristic diversity. West-to-east gradients in rainfall generate sharp spatial variation in species assemblages and vegetation types (Monks *et al.* 2019). The coexistence of high species richness and elevated rarity presents substantial challenges for conservation planning. Localized super-hotspots of biodiversity require accurate prediction models with sufficient spatial resolution to guide habitat prioritization, ecological restoration and targeted interventions for threatened taxa.

The biodiversity dataset used in this study includes 4989 sample observations of vascular plant species richness across SWAFR (Mokany *et al.* 2022), as shown in Figure 2. Observed richness values range from 0.4 to 135.8, with a mean of 41.1. Each data point represents vascular plant species richness within a 400 m² circular plot, where at least 70% of the recorded species are native to Australia and at least 90% are taxonomically identified to the species level, ensuring the ecological representativeness of the samples. To prepare the data for spatial modelling and prediction, all observations were aggregated into 10 km resolution grid cells, resulting in a total of 958 spatial units with aggregated species richness values. The resulting 10-km dataset was used for model training and evaluation.

3.2. Explanatory variables

Twelve explanatory variables are collected to capture the spatial patterns of environmental, climatic and edaphic conditions influencing vascular plant species richness. These variables are categorized into three groups: geography (elevation and slope), climate and environment (precipitation, shortwave radiation, distance to water, and

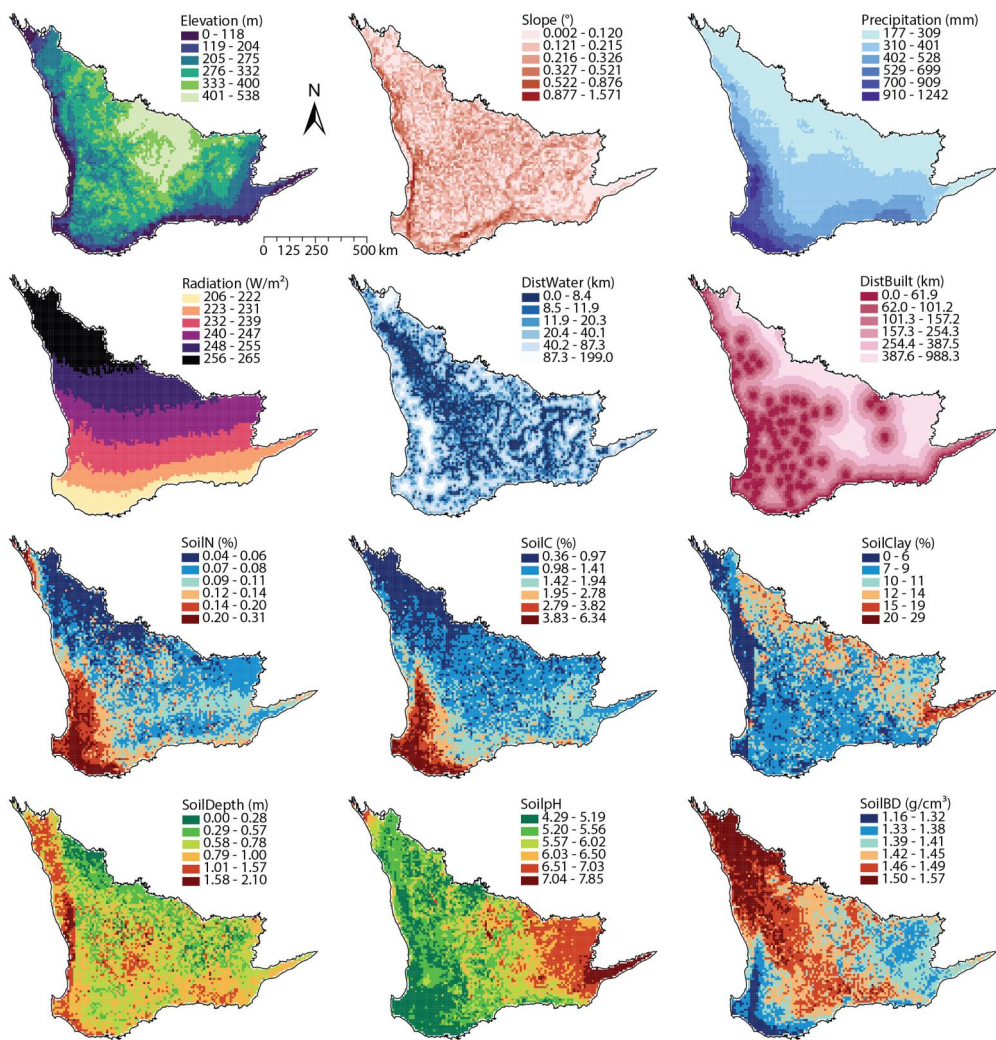


Figure 3. Spatial distributions of variables used for the spatial prediction of species richness.

distance to built-up areas), and soil properties (nitrogen content, organic carbon, clay fraction, depth, pH and bulk density). Descriptions of explanatory variables are presented in [Table 3](#). Geographic and climatic variables were acquired from global datasets processed using the Google Earth Engine platform. Elevation and slope data were collected from the Shuttle Radar Topography Mission (SRTM) dataset with a spatial resolution of 30 m and processed across Australia (Farr *et al.* 2007). Precipitation data were obtained from the Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) dataset, accumulated over 2013–2023 (Funk *et al.* 2015), and shortwave radiation data were obtained from the Famine Early Warning Systems Network (FEWS NET) Land Data Assimilation System (FLDAS) (McNally *et al.* 2017). The distance-based environmental variables were derived from the 2015 national land cover classification with a spatial resolution of 30 m published by Digital Earth Australia (Owers *et al.* 2022). Distance to water was calculated by measuring the shortest distance from each

grid cell to the nearest inland water body, salt lake or ocean, based on the land cover water class. Distance to built-up areas was defined as the shortest distance to pixels classified as artificial surfaces or urban settlements. Soil variables were derived from the CSIRO TERN Landscape Phase II dataset with a spatial resolution of 90 m released in 2024, representing averaged topsoil conditions (0–5 cm) over the period 1950–2023 (Malone and Searle 2022). Figure 3 shows the spatial distributions of the three categories of explanatory variables across the study region. The spatial gradients in geographical, climate, environmental and soil properties capture the geoclimatic diversity of the SWAFR, forming a multidimensional covariate space supporting species richness modelling and enabling spatial prediction sensitive to local environmental variation.

3.3. Experiment design

3.3.1. Data pre-processing

Before applying GCF to biodiversity prediction, the explanatory variables were preprocessed to ensure spatial consistency with the response variable. Species richness was represented on a 10-km grid, and all explanatory covariates, including geographic, climatic, environmental and soil variables, were resampled to this resolution. As a descriptive check, Pearson's correlation coefficients were calculated between each explanatory variable and species richness across the observed locations. All 12 explanatory variables showed significant correlations with species richness and were retained for subsequent modelling. This correlation check was used to summarize the empirical relevance of the candidate covariates, rather than as a response-driven feature selection step.

3.3.2. GCF modelling of biodiversity

GCF modelling of the biodiversity prediction consists of four steps. In the first step, the 11 spatial pattern operators listed in Table 1 are applied to each covariate over multiscale neighbourhoods defined by buffer radii $b = \{20, 30, \dots, 100\}$ km, with the normalization radius $r = 100$ km for ψ_L . This produces the spatial pattern features $\psi(x)$, which describe local spatial dependence, local heterogeneity, geocomplexity, multiscale variation and outlyingness around each location.

In the second step, neighbourhood distribution features are generated over the same nine buffer radii using quantile levels $\tau = \{0, 0.05, \dots, 1\}$. For each covariate, this gives $9 \times 21 = 189$ neighbourhood distribution features; across the 12 covariates, this gives $12 \times 9 \times 21 = 2268$ features in total. These features are combined with the original predictors and the spatial pattern features in the GCF feature vector $\Phi(s)$.

In the third step, variable selection is performed using the two-stage procedure. Functional reduction uses a fine scale band $\{20, 30\}$ km and a broad scale band $\{90, 100\}$ km, reducing the full candidate set to 372 features by retaining, for each covariate, five distributional functionals and band-averaged spatial pattern operators. Spatial-block stability selection then partitions the study region into square blocks of side length $L = 2a$, where a is the residual-variogram range, so that each block exceeds the estimated spatial dependence length. The selection uses $M = 80$ spatial-block resampling rounds. In each round, a fraction $\gamma = 0.7$ of blocks is sampled, a

200-tree RF is fitted, and the top $k = 20$ derived features ranked by impurity importance are marked as selected. Groups defined by covariate and feature category are retained when their selection frequency is at least $\pi = 0.6$, and each retained group contributes one representative feature.

In the fourth step, spatial learning is performed by fitting a 500-tree RF model to the selected GCF predictors. RF constructs an ensemble of decision trees trained on bootstrap samples and aggregates their predictions to improve generalization and reduce variance (Breiman 2001). Each tree partitions the predictor space by axis-aligned splits and fits a separate constant within each region, so RF imposes no monotonicity constraint and represents non-monotonic covariate–response relationships (Hastie *et al.* 2009, Scornet *et al.* 2015).

3.3.3. Model validation and spatial prediction

Model validation evaluates the biodiversity prediction across four components, contrasting the GCF representation with the original covariates. Variable selection is re-estimated within each training fold throughout validation, ensuring that validation response information does not influence variable selection. The adopted RF model is benchmarked against six comparison learners, each configured with independently selected hyperparameters. Cubist combines 20 committees with a 9-neighbour instance correction; GLMNET fixes $\alpha = 0$ with the penalty λ chosen by fivefold cross-validation; LM uses ordinary least squares; XGBoost builds depth-3 trees with a learning rate of 0.03 over 1000 rounds, an L_2 regularization of $\lambda = 3$, and row and column subsampling of 0.8 and 0.5; the SVM uses a radial basis kernel with cost 0.5; and the KNN model uses $k = 20$ under the optimal kernel.

The first aspect assesses prediction accuracy using fivefold random and spatial-block cross-validation. The random cross-validation assigns sampled locations to folds at random and measures accuracy at validation locations drawn from the same spatial distribution as the training data. The spatial-block cross-validation divides the study region into contiguous square blocks of side length L and assigns whole blocks to folds randomly to separate training and validation samples and reduce short-range spatial dependence between them. For each evaluated learner, performance is reported under both cross-validation schemes for the covariate-only and GCF models using coefficient of determination (R^2) and RMSE.

The second aspect isolates the contribution of the GCF feature components. Four predictor sets are compared under spatial-block cross-validation: the covariate-only set x , the pattern-augmented set $x + \psi$, the distribution-augmented set $x + Z_x$, and the full GCF representation. The RMSE reduction relative to the covariate-only model is then decomposed into a spatial-pattern contribution, a neighbourhood-distribution contribution, and a shared term representing overlap or complementarity between the two derived feature components.

The third aspect evaluates robustness under data scarcity. Sampling-density experiments vary the training fraction and track both prediction gain and between-fold variability. Spatial-extrapolation experiments withhold entire geographic regions and vary the spatial-block side length to test whether the GCF gain transfers to unsampled space. Covariate-scarcity experiments remove covariates either from strongest to

weakest or at random, measuring how prediction accuracy degrades as explanatory information is withdrawn.

The fourth aspect maps species richness across the full prediction grid using the RF model refitted on all sampled locations. The resulting GCF prediction surface is compared with the covariate-only surface. The local variance (V_l) of each predicted surface is used to assess local spatial variability and surface stability.

4. Results

4.1. Data pre-processing

To evaluate the ecological relevance of explanatory variables for biodiversity prediction, Pearson's correlation coefficients were computed between species richness and each of the 12 environmental covariates. The results, shown in Figure 4, indicate that all variables are significantly correlated with species richness at the $p < 0.01$ level. Among them, the top three variables exhibiting the strongest correlations were precipitation ($r = 0.464$, $p < 0.01$), distance to built-up land ($r = -0.454$, $p < 0.01$) and soil total nitrogen ($r = 0.399$, $p < 0.01$). The correlation analysis confirms the ecological relevance of the explanatory variables and supports their inclusion in GCF.

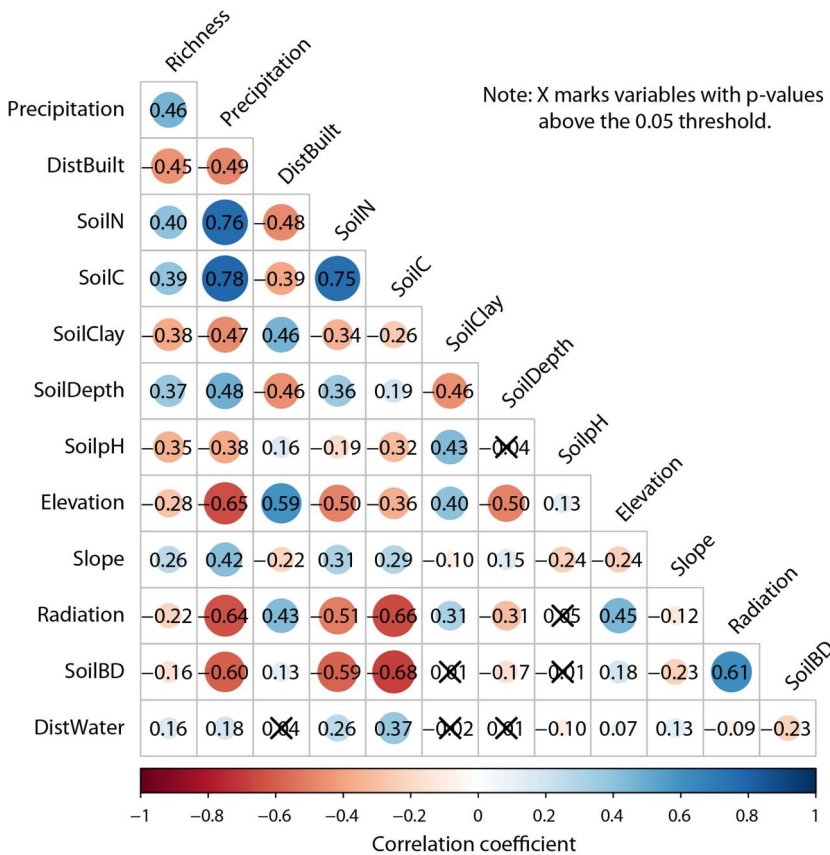


Figure 4. Correlation coefficients between species richness and the explanatory variables.

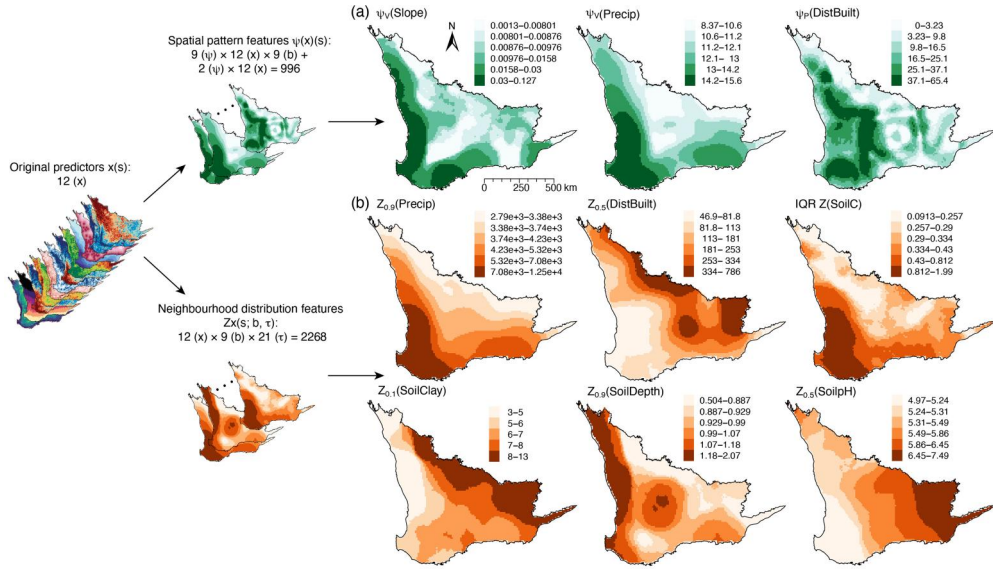


Figure 5. Spatial distributions of representative predictors used in the GCF model, including (a) spatial pattern features derived from original variables and (b) neighbourhood distribution features summarizing original variables over neighbourhoods of the prediction grid. The left schematic shows the expansion of the 12 original predictors into two derived feature categories. The mapped predictors are the nine derived features retained by the full-sample variable selection. ψ_V = log local variance; ψ_P = positive z-outlier strength; Z_τ = neighbourhood τ -quantile, with $\text{IQR } Z = Z_{0.75} - Z_{0.25}$.

4.2. GCF-based biodiversity prediction

4.2.1. Spatial pattern features

The first step of GCF-based biodiversity prediction generates spatial pattern features that quantify local spatial structure in the explanatory variables. The 11 operators enhance the input data by revealing spatial structure not explicitly represented in the original variables. Figure 5(a) maps the three spatial pattern features retained by the variable selection for the final prediction model: the log local variance (ψ_V) of slope and of precipitation, and the positive z-outlier strength (ψ_P) of distance to built-up areas, all computed at the broad neighbourhood scale with neighbourhoods of 90–100 km. Compared to the original variable values, the spatial pattern features reveal finer-scale spatial variation and regions of higher local heterogeneity, variability and outlyingness.

4.2.2. Neighbourhood distribution features

Neighbourhood distribution features summarize the distribution of covariate values within multiscale neighbourhoods around each location. Figure 5(b) presents the six neighbourhood distribution features retained by the variable selection at the broad scale of 90–100 km: the 90th quantile of precipitation ($Z_{0.90}$), the median distance to built-up areas ($Z_{0.50}$), the interquartile range of soil organic carbon ($\text{IQR } Z = Z_{0.75} - Z_{0.25}$), the 10th quantile of soil clay ($Z_{0.10}$), the 90th quantile of soil depth ($Z_{0.90}$) and the median soil pH ($Z_{0.50}$). These features are generated by applying quantile-based summaries to

Table 4. Summary of the three predictor categories used in GCF-based prediction.

Predictor category	Number of variables
Original predictors $x(s)$	12 (x) = 12
Spatial pattern features $\psi(x)(s)$	$9 (\psi) \times 12 (x) \times 9 (b) + 2 (\psi) \times 12 (x) = 996$
Neighbourhood distribution features $Z_x(s; b, \tau)$	$12 (x) \times 9 (b) \times 21 (\tau) = 2268$
Total candidate predictors in GCF	$12 + 996 + 2268 = 3276$

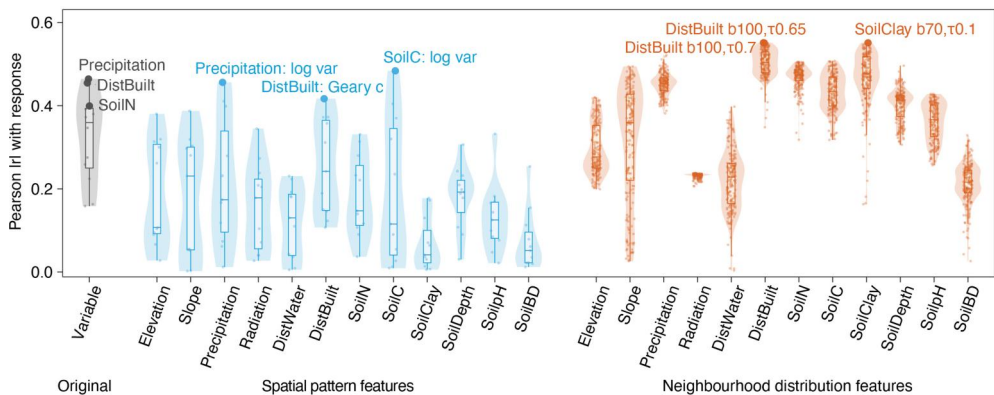


Figure 6. Absolute Pearson's correlation between species richness and GCF-derived candidate predictors using samples.

covariate values within a buffer of radius b around each prediction location v . In the biodiversity case, each original variable produces $9 \times 21 = 189$ neighbourhood distribution candidates from nine buffer radii and 21 quantile levels, which are then reduced to compact distributional functionals over fine and broad scale bands. These features capture neighbourhood-level distributional context beyond point covariate values. For example, $Z_{0.90}$ of precipitation highlights areas influenced by high-rainfall neighbourhoods, while IQR Z of soil organic carbon identifies zones with strong local variability.

4.2.3. Variable selection and spatial modelling

In the biodiversity prediction, GCF incorporates three predictor categories: original predictors x , spatial pattern features $\psi(x)$ and neighbourhood distribution features $Z_x(s; b, \tau)$. Their counts are summarized in Table 4.

Figure 6 presents the absolute Pearson correlations between species richness and GCF-derived predictors, grouped by predictor category. Among the original covariates, precipitation shows the strongest correlation with richness ($|r| = 0.464$), followed by distance to built-up areas ($|r| = 0.455$) and soil nitrogen ($|r| = 0.399$). Neighbourhood distribution features show the strongest individual associations with species richness: the median of distance to built-up areas and the 10th quantile of soil clay both reach $|r| = 0.544$, about 17% higher than the absolute correlation of precipitation, with several additional distribution features exceeding $|r| = 0.50$. Spatial pattern features have a lower typical correlation, with a median $|r|$ of approximately 0.13, but include strongly correlated features, such as the local variance of precipitation ($|r| = 0.494$) and soil organic carbon ($|r| = 0.492$), both exceeding all original covariates. Overall, neighbourhood distribution features achieve the highest maximum correlation ($|r| = 0.544$) and a median correlation ($|r| = 0.348$) comparable to that of the original

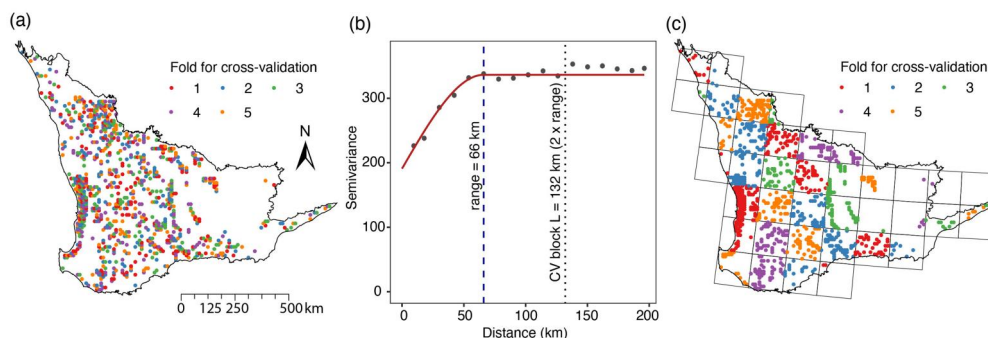


Figure 7. Fivefold cross-validation schemes for biodiversity modelling. (a) Random cross-validation folds, (b) empirical variogram of species richness used to guide the spatial-block size and (c) spatial-block cross-validation folds.

covariates ($|r| = 0.359$), while spatial pattern features contribute fewer but highly informative descriptors.

To improve interpretability and computational efficiency, functional reduction followed by spatial-block stability selection is applied to obtain a compact predictor subset for RF modelling. Applied to the full dataset for the final prediction model, the two-stage procedure retains nine of the 24 groups defined by covariate and derived-feature category, and the final predictor set comprises the 12 original covariates together with the nine representative derived features mapped in Figure 5.

4.3. Model validation and spatial prediction

4.3.1. Random and spatial cross-validation

Model accuracy is assessed using two complementary fivefold cross-validation schemes, shown in Figure 7. In the random cross-validation, the 958 aggregated biodiversity observations are assigned to five folds of roughly equal size, with about 192 samples per fold. In the spatial-block cross-validation, the samples are grouped into 38 occupied square blocks of side length $L = 132$ km, and whole blocks are assigned to five spatially disjoint folds so that training and validation samples do not occur within the same block. The block size is determined from the residual variogram of species richness after removing the covariate trend (Figure 7(c)), which gives a spatial-dependence range of $a \approx 66$ km. The block side length is therefore set to $L = 2a = 132$ km, so that validation blocks are separated beyond the estimated short-range dependence.

Table 5 shows the results of the fivefold random and spatial-block cross-validation. First, every learner improves under GCF relative to its covariate-only counterpart in both cross-validation schemes. Under random cross-validation, GCF increases R^2 by 0.7–13.2% and reduces RMSE by 0.3–3.6%; under spatial-block cross-validation, it increases R^2 by 8.2–21.1% and reduces RMSE by 1.6–3.5%. Second, RF is the most accurate learner under both cross-validation schemes with the highest R^2 and the lowest RMSE. Under spatial-block cross-validation, its GCF model achieves $R^2 = 0.346$ and

Table 5. Fivefold random and spatial-block cross-validation results for biodiversity modelling.

Model	Random CV						Spatial CV					
	R^2			RMSE			R^2			RMSE		
	Baseline	GCF	Imp. (%)	Baseline	GCF	Imp. (%)	Baseline	GCF	Imp. (%)	Baseline	GCF	Imp. (%)
RF	0.441	0.444	+0.7%	17.114	17.043	+0.4%	0.296	0.346	+16.7%	17.966	17.428	+3.0%
Cubist	0.431	0.435	+0.9%	17.230	17.181	+0.3%	0.283	0.342	+21.1%	18.065	17.506	+3.1%
KNN	0.427	0.432	+1.3%	17.321	17.232	+0.5%	0.296	0.320	+8.2%	18.039	17.743	+1.6%
XGBoost	0.401	0.428	+6.9%	17.704	17.291	+2.3%	0.261	0.315	+20.3%	18.304	17.795	+2.8%
SVM	0.398	0.419	+5.4%	17.750	17.418	+1.9%	0.275	0.327	+18.6%	18.489	17.849	+3.5%
GLMNET	0.350	0.393	+12.3%	18.451	17.822	+3.4%	0.259	0.302	+16.9%	18.575	18.054	+2.8%
LM	0.349	0.395	+13.2%	18.464	17.801	+3.6%	0.253	0.286	+13.0%	18.647	18.279	+2.0%

Baseline and GCF denote the covariate-only and GCF models, respectively, Imp. (%) denotes the relative improvement from the covariate-only model to GCF, and bold values indicate the best cross-validation result in each column.

RMSE = 17.428, compared with $R^2 = 0.296$ and RMSE = 17.966 for the covariate-only model, and is therefore adopted for the case study. Third, accuracy is consistently lower under spatial-block cross-validation than under random cross-validation, reflecting the removal of short-range spatial dependence between training and validation samples. Under this stricter cross-validation, the larger GCF gains are consistent with improved transferability to spatially separated and unsampled areas.

4.3.2. Contributions of GCF features

Beyond model accuracy, a feature-set comparison is used to isolate the contribution of each GCF-derived feature component under the spatial-block cross-validation. Table 6 compares four predictor sets for the RF: the covariate-only set x , the pattern-enhanced set $x + \psi$, the neighbourhood-distribution set $x + Z_x$ and the full GCF representation. The covariate-only model attains $R^2 = 0.296$ and RMSE = 17.966. Adding spatial pattern features improves RMSE to 17.500, while adding neighbourhood distribution features improves RMSE to 17.613. The full GCF representation achieves the best performance, with $R^2 = 0.346$ and RMSE = 17.428. These results show that both derived feature blocks provide transferable gains under spatial-block cross-validation, and that their combination gives the strongest improvement. The four predictor sets also demonstrate the effectiveness of GCF in unifying existing feature constructions: covariates alone (x), covariates with spatial pattern features ($x + \psi$) and covariates with neighbourhood distribution features ($x + Z_x$) are all nested within the full GCF representation.

In addition, Table 7 presents the Shapley decomposition of spatial-block R^2 for the RF model. The covariates contribute approximately 45.6%, the spatial pattern features contribute 24.8%, and the neighbourhood distribution features contribute 29.6%. Together, the two GCF-derived features account for more than half of the explained variance, indicating that the added spatial-context features make a substantial contribution beyond the original covariates.

Table 6. Feature-set comparison for biodiversity modelling under spatial-block cross-validation using random forest.

Feature set	R^2	RMSE	$\Delta R^2\%$	Δ RMSE%
x	0.296	17.966	–	–
$x + \psi$	0.337	17.500	+13.7%	+2.6%
$x + Z_x$	0.329	17.613	+11.2%	+2.0%
GCF	0.346	17.428	+16.7%	+3.0%

ΔR^2 (%) and Δ RMSE (%) denote the relative changes from the covariate-only set x to each feature set.

Table 7. Spatial-block R^2 Shapley decomposition for the biodiversity modelling using random forest.

Category	Shapley ϕ_c	Share
x (covariates)	0.158	45.6%
ψ (spatial pattern)	0.086	24.8%
Z_x (neighbourhood distribution)	0.102	29.6%
Sum of Shapley values $\sum_c \phi_c$	0.346	100%

ϕ_c is the Shapley contribution of each predictor category to the GCF R^2 .

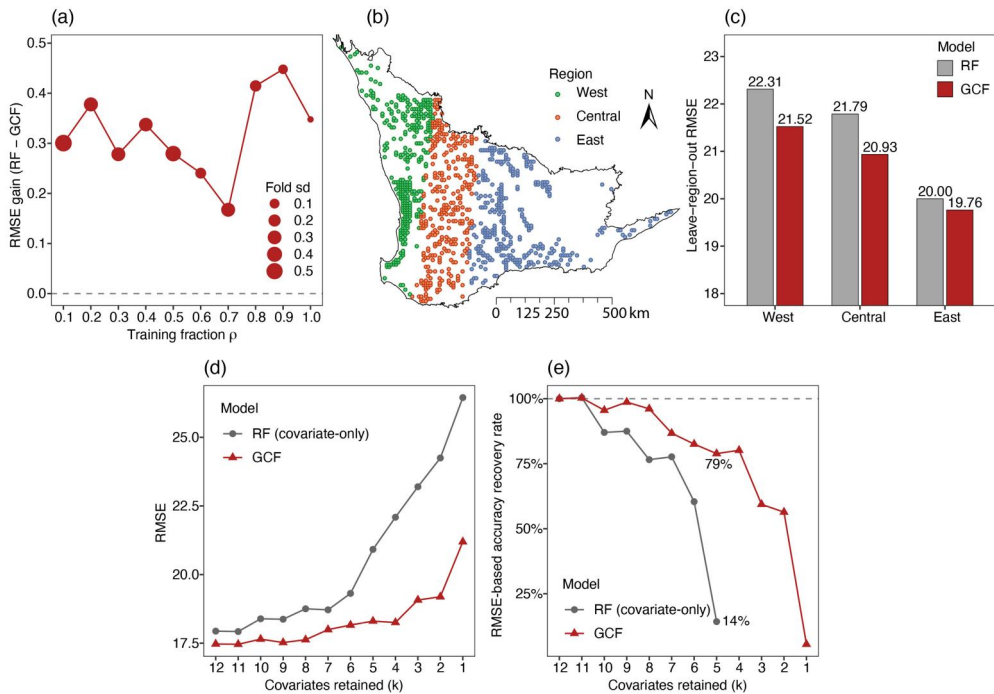


Figure 8. Robustness analysis under sparse sampling, spatial extrapolation and covariate scarcity. (a) Training-density RMSE gain, defined as the RMSE of the covariate-only RF (denoted RF) minus that of GCF, with point size indicating the between-fold standard deviation. (b) West-central-east regional split. (c) Leave-region-out RMSE for the covariate-only RF and GCF models. (d) RMSE and (e) RMSE-based accuracy recovery rate (ARR) under covariate scarcity, $ARR = (RMSE_{mean} - RMSE(k)) / (RMSE_{mean} - RMSE_{k=12})$ as the strongest covariates are removed first from $k = 12$ to $k = 1$, where $RMSE_{mean} = sd(y)$, the RMSE of predicting the global mean.

4.3.3. Robustness to sparse sampling, spatial extrapolation and covariate scarcity

Figure 8 evaluates the robustness of the GCF gain from three perspectives. First, under sparse sampling, the RMSE gain remains positive across all training fractions, increasing from 0.30 at $p = 0.1$ –0.45 at $p = 0.9$, while the between-fold spread decreases as more training data become available (Figure 8(a)). Second, the leave-region-out experiment shows that the gain transfers to spatially withheld regions: GCF reduces RMSE by 0.85 in the central block and 0.79 in the western block, with a smaller but still positive gain of 0.24 in the eastern block. In all three withheld regions, GCF gives lower RMSE than the covariate-only RF (Figure 8(b,c)). Third, under covariate scarcity, GCF degrades more gradually than RF when the strongest covariates are removed first. For example, when the number of retained covariates is reduced from 12 to 5, the GCF recovers 79% of the full-covariate prediction capacity whereas RF recovers only 14%. With only one covariate retained, GCF RMSE increases by 21%, compared with 47% for RF, and GCF maintains positive R^2 throughout, whereas RF fails to maintain positive R^2 when fewer than four covariates remain (Figure 8(d,e)).

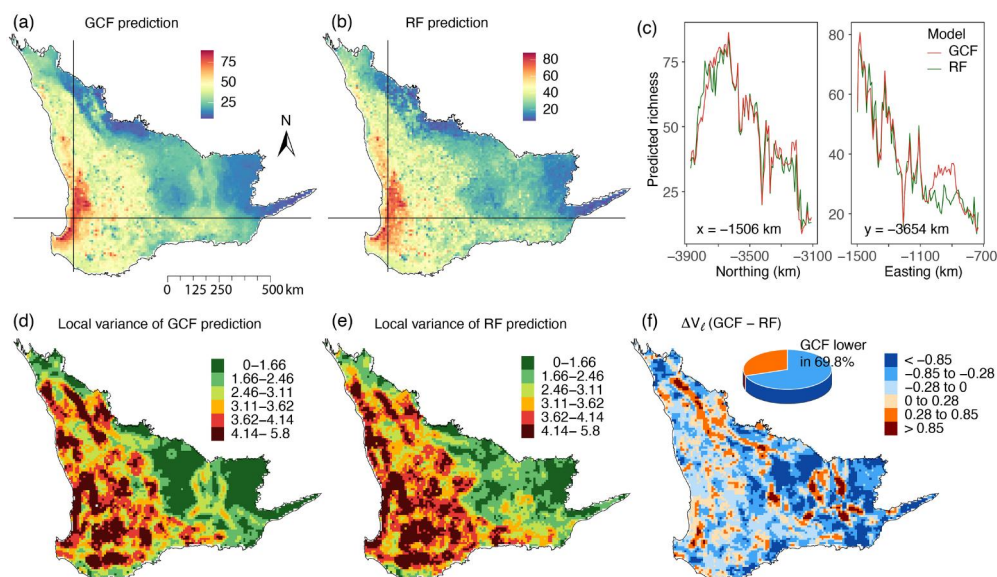


Figure 9. Spatial prediction and evaluation of biodiversity using the GCF model. (a) GCF-based biodiversity prediction; (b) covariate-only random forest (RF)-based biodiversity prediction; (c) comparison of GCF and RF predictions along two cross-sections at $x = -1506$ km and $y = -3654$ km; (d) local variance (V_L) of the GCF prediction; (e) local variance (V_L) of the RF prediction; (f) spatial difference in local variance (ΔV_L) between the GCF and RF predictions.

4.3.4. Spatial prediction

To evaluate the spatial characteristics of the prediction outputs, spatial prediction maps and their local-structure analyses are shown in Figure 9. Figure 9(a,b) displays the predicted vascular plant species richness from the GCF-based RF model and the covariate-only RF model, respectively. The GCF prediction surface shows smoother spatial distributions and more coherent transitions across the study area, whereas the covariate-only RF surface appears more fragmented. The cross-sectional plots in Figure 9(c), taken along transects at $x = -1506$ km and $y = -3654$ km, further show that GCF preserves stronger spatial contrasts, including both high- and low-richness values, while the RF model tends to dampen local peaks. Figure 9(d,e) shows the local variance V_L of the GCF and RF predictions, where V_L is calculated as the variance of predicted values within each location's neighbourhood. Figure 9(f) maps the difference in local variance, ΔV_L . GCF has lower local variance across 69.8% of the region, indicating greater local stability and spatial coherence in the predicted surface. Overall, GCF improves not only prediction accuracy but also the spatial coherence and local stability of the predicted biodiversity surface.

5. Discussion

This study developed a GCF model to address the underuse of the spatial structure embedded in covariates in spatial prediction. GCF expands each covariate into spatial pattern features and neighbourhood distribution features to improve spatial prediction models without increasing the number of response observations. The simulation and

biodiversity prediction studies show that GCF improved all evaluated learners over their covariate-only counterparts under both random and spatial-block cross-validation. The spatial-block results showed consistent gains in prediction performance, and the neighbourhood distribution features contributed strongly to spatially transferable accuracy. The spatial predictions were more spatially coherent than those from the covariate-only models, while preserving local variability and spatial contrasts. Thus, spatial prediction can be improved by representing the geographical characteristics of covariates, not only the point observation values. GCF also effectively operationalizes the primacy of data principle in geospatial modelling that improvements can arise from richer spatial representation of covariates.

The improved performance of GCF stems from redefining the input structure of spatial models. Rather than relying solely on covariate values or increasing model complexity, GCF expands the feature space through two complementary operators. The spatial pattern operator measures local structures of spatial autocorrelation, local heterogeneity, geocomplexity and local outlyingness, and the neighbourhood distribution operator summarizes the distribution of covariate values within multiscale neighbourhoods. The variable selection strategy combines response-independent functional reduction with spatial-block stability selection to retain a compact set of spatially transferable and interpretable features, such as high quantiles for local covariate extremes and interquartile ranges for local variability.

Generalization of GCF to new populations and sampling conditions follows mainly from two conditions: a response that depends on geographical context, and covariate support that extends beyond the response samples. These conditions define settings in which GCF is expected to provide greater benefits, and the expected gains increase as the response sample becomes sparser relative to the covariate support, because neighbourhood features supply contextual information beyond that available at the sampled locations. Two experiments support the expectation: under covariate scarcity, GCF recovered 79% of the full-covariate prediction capacity with five covariates against 14% for the covariate-only RF, and under leave-region-out validation the gain transferred to all three withheld regions. Model improvements are correspondingly smaller where response sample density approaches the covariate resolution, so that neighbourhood summaries duplicate information already in x , or where the process scale falls outside the range spanned by the buffer radii.

This study contains the following limitations. First, expanding covariates to include spatial pattern and neighbourhood distribution features creates a high-dimensional candidate space and requires variable selection. Second, the current implementation uses predefined buffer radii and quantile levels, which may not be optimal for all spatial processes, study regions or resolutions. In addition, the operators implemented here describe local heterogeneity, so populations dominated by stratified heterogeneity would call for stratum-based operators within the same construction. Geographical detector models are useful for quantifying spatial associations under stratified heterogeneity (Wang *et al.* 2010), while further development of stratum-based operators for individual variables would be valuable. Third, GCF was evaluated here for static spatial prediction, and further work is needed to test its behaviour in temporal and spatiotemporal settings (Lyu *et al.* 2026b) and in

causal analysis of spatial cross-sectional data (Lyu *et al.* 2026a). Future studies could improve neighbourhood distribution features using other distributional summaries such as means, upper-tail statistics, threshold exceedances and entropy, and integrate GCF with deep learning, transfer learning and foundation models to broaden the applicability of GCF and strengthen data-centric spatial modelling in geospatial analytics.

6. Conclusions

This study developed the GCF model to improve spatial prediction by enriching each covariate with spatial pattern and neighbourhood distribution features defined around each location. GCF formalizes a data-centric approach in GIScience by using the spatial structure embedded in covariates, including spatial dependence, local heterogeneity, geocomplexity, multiscale variation and local outlyingness, rather than relying only on point covariate values. The simulation and biodiversity prediction studies showed that GCF improved prediction accuracy, spatial generalizability and spatial coherence under both random and spatial-block cross-validation. The results show that richer spatial representation of covariates can enhance spatial learning across different model types and complex geospatial contexts. The variable selection strategy further retains a compact set of spatially transferable and interpretable features. Future work should address the high dimensionality of the candidate feature space and develop adaptive neighbourhood operators and extensions to temporal and spatiotemporal prediction tasks. As a general spatial representation model, GCF can be applied to other spatial prediction problems where the spatial characteristics of covariates encode important geographical, environmental or urban-process information.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Author contributions

CRediT: **Yongze Song**: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Data and codes availability statement

The data and codes supporting the findings of the study are available on Figshare at <https://doi.org/10.6084/m9.figshare.29425739> and the GitHub repository at <https://github.com/yongze-song/gcf>.

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